705.641.81: Homework 1

David Carrillo

1 Linear Algebra Review

1.1 Basic Operations

With $\alpha = 2$, $\mathbf{x} = [0, 1, 2]^\top$, $\mathbf{y} = [3, 4, 5]^\top$, $\mathbf{z} = [1, 2, -1]^\top$, and $A = \begin{bmatrix} 3 & 2 & 2 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$:

1. $\sum_{i=1}^n x_i y_i = (0)(3) + (1)(4) + (2)(5) = 0 + 4 + 10 = \mathbf{14}$.

2. $\sum_{i=1}^n x_i z_i = (0)(1) + (1)(2) + (2)(-1) = 0 + 2 - 2 = \mathbf{0}$. The inner product vanishes, which is what makes $\mathbf{x}$ and $\mathbf{z}$ orthogonal.

3. $\alpha(\mathbf{x} + \mathbf{y}) = 2[0 + 3,\ 1 + 4,\ 2 + 5]^\top = 2[3, 5, 7]^\top = [6, 10, 14]^\top$.

4. $\|\mathbf{x}\| = \sqrt{0^2 + 1^2 + 2^2} = \sqrt{5} \approx \mathbf{2.2361}$.

5. $\mathbf{x}^\top = \begin{bmatrix} 0 & 1 & 2 \end{bmatrix}$, the 1×3 row vector obtained from the 3×1 column $\mathbf{x}$.

6. $A\mathbf{x}$: each entry is a row of A dotted with $\mathbf{x}$,

$$A\mathbf{x} = \begin{bmatrix} (3)(0) + (2)(1) + (2)(2) \\ (1)(0) + (3)(1) + (1)(2) \\ (1)(0) + (1)(1) + (3)(2) \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \\ 7 \end{bmatrix}.$$

7. $\mathbf{x}^\top A\mathbf{x} = \mathbf{x} \cdot (A\mathbf{x}) = (0)(6) + (1)(5) + (2)(7) = 0 + 5 + 14 = \mathbf{19}$.

1.2 Matrix Algebra Rules

Here $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ are $n \times 1$ column vectors and $\{A, B, C\}$ are $n \times n$ real matrices.

1. $\mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i$. **True.** This is the definition of the inner product: the $(1 \times n)(n \times 1)$ product is the 1×1 sum of elementwise products.

2. $\mathbf{x}^\top \mathbf{x} = \|\mathbf{x}\|^2$. **True.** Setting $\mathbf{y} = \mathbf{x}$ above gives $\sum_i x_i^2$, which is the definition of the squared Euclidean norm.

3. $\mathbf{x}^\top \mathbf{x} = \mathbf{x}\mathbf{x}^\top$. **False.** The shapes do not even match: $\mathbf{x}^\top \mathbf{x}$ is 1×1 while $\mathbf{x}\mathbf{x}^\top$ is $n \times n$. For $\mathbf{x} = [1, 2, 3]^\top$, $\mathbf{x}^\top \mathbf{x} = 14$ but $\mathbf{x}\mathbf{x}^\top = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$, a rank-one matrix.

4. $(\mathbf{x}-\mathbf{y})^\top(\mathbf{y}-\mathbf{x}) = \|\mathbf{x}\|^2 - 2\mathbf{x}^\top\mathbf{y} + \|\mathbf{y}\|^2$. **False**, and the error is a sign. Since $\mathbf{y}-\mathbf{x} = -(\mathbf{x}-\mathbf{y})$,

$$(\mathbf{x}-\mathbf{y})^\top(\mathbf{y}-\mathbf{x}) = -(\mathbf{x}-\mathbf{y})^\top(\mathbf{x}-\mathbf{y}) = -\|\mathbf{x}-\mathbf{y}\|^2 = -(\|\mathbf{x}\|^2 - 2\mathbf{x}^\top\mathbf{y} + \|\mathbf{y}\|^2).$$

The left side is ≤ 0 always, while the right side is ≥ 0 always. They agree only when $\mathbf{x} = \mathbf{y}$. With $\mathbf{x} = [1,2,3]^\top$ and $\mathbf{y} = [0,1,0]^\top$ the left side is -11 and the right side is $+11$. The correct identity uses $(\mathbf{x}-\mathbf{y})^\top(\mathbf{x}-\mathbf{y})$.

5. $AB = BA$. **False.** Matrix multiplication does not commute. With $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $AB = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ but $BA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$.

6. $A(B+C) = AB + AC$. **True.** Matrix multiplication distributes over addition, since $\sum_k a_{ik}(b_{kj} + c_{kj}) = \sum_k a_{ik}b_{kj} + \sum_k a_{ik}c_{kj}$.

7. $(AB)^\top = A^\top B^\top$. **False.** The transpose reverses the order: $(AB)^\top = B^\top A^\top$. With the same A and B as above, $(AB)^\top = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ and $A^\top B^\top = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$, which differ, while $B^\top A^\top = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ matches.

8. $\mathbf{x}^\top A\mathbf{y} = \mathbf{y}^\top A^\top \mathbf{x}$. **True.** The quantity $\mathbf{x}^\top A\mathbf{y}$ is 1×1, so it equals its own transpose, and $(\mathbf{x}^\top A\mathbf{y})^\top = \mathbf{y}^\top A^\top \mathbf{x}$ by reversing the order of the three factors.

9. $A^\top A = I$ is the columns of A are orthonormal. **True.** Writing $A = [\mathbf{a}_1 \cdots \mathbf{a}_n]$ by columns, the (i,j) entry of $A^\top A$ is $\mathbf{a}_i^\top \mathbf{a}_j$. So $A^\top A = I$ says exactly that $\mathbf{a}_i^\top \mathbf{a}_j = 0$ for $i \neq j$ and $\mathbf{a}_i^\top \mathbf{a}_i = 1$, which is the definition of an orthonormal set of columns. The implication runs both ways.

1.3 Matrix operations

Let $B = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$.

Is B invertible?

No. Expanding the determinant along the first row,

$$\det B = 1[(2)(1) - (-1)(-1)] - (-1)[(-1)(1) - (-1)(0)] + 0 = 1(2-1) + 1(-1) = 1 - 1 = 0.$$

A zero determinant means B is singular, so B^{-1} does not exist. The reason is visible in the structure: every row sums to zero, so $B[1,1,1]^\top = \mathbf{0}$ and the vector $[1,1,1]^\top$ spans the null space. The rank is 2.

Is B diagonalizable?

Yes. The characteristic polynomial is

$$\det(B - \lambda I) = (1-\lambda)[(2-\lambda)(1-\lambda) - 1] + 1[-(1-\lambda)] = (1-\lambda)[(2-\lambda)(1-\lambda) - 2],$$

and since $(2-\lambda)(1-\lambda) - 2 = \lambda^2 - 3\lambda = \lambda(\lambda - 3)$,

$$\det(B - \lambda I) = (1-\lambda)\lambda(\lambda - 3), \qquad \text{so} \qquad \lambda \in \{0, 1, 3\}.$$

Three distinct eigenvalues in a 3×3 matrix give three linearly independent eigenvectors, so B is diagonalizable. It is also real symmetric, so the spectral theorem guarantees this independently. Solving $(B - \lambda I)\mathbf{v} = \mathbf{0}$ for each eigenvalue:

$$\lambda_1 = 0: \ \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \qquad \lambda_2 = 1: \ \mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \qquad \lambda_3 = 3: \ \mathbf{v}_3 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}.$$

The diagonalization is $B = PDP^{-1}$ with

$$P = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & -2 \\ 1 & 1 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Because B is symmetric its eigenvectors are mutually orthogonal, so normalizing the columns of P gives an orthogonal Q with $B = QDQ^\top$:

$$Q = \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -2/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix}.$$

2 Taking Chances: Probability Review

2.1 Basic probability

1. Two dice give $6 \times 6 = 36$ equally likely ordered outcomes. Counting by absolute difference $d = |d_1 - d_2|$, there are $12 - 2d$ outcomes for each $d \geq 1$, so $d = 3$ gives 6 outcomes, $d = 4$ gives 4, and $d = 5$ gives 2. The winning event $d \geq 3$ therefore has $6 + 4 + 2 = 12$ outcomes:

$$\mathbb{P}(\text{win}) = \frac{12}{36} = \frac{1}{3}.$$

The payoff is \$15 on a win and \$0 otherwise, so $\mathbb{E}[\text{payoff}] = 15 \cdot \frac{1}{3} + 0 \cdot \frac{2}{3} = \$5$. A fair ticket price is **\$5**, the price at which the game has zero expected profit.

2. From $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A, B)$ with $\mathbb{P}(A, B) = 0$:

$$0.95 = 0.4 + \mathbb{P}(B) - 0, \qquad \text{so} \qquad \mathbb{P}(B) = \mathbf{0.55}.$$

3. If A and B are independent then $\mathbb{P}(A, B) = \mathbb{P}(A)\mathbb{P}(B) = 0.4\mathbb{P}(B)$, so

$$0.95 = 0.4 + \mathbb{P}(B) - 0.4\mathbb{P}(B) = 0.4 + 0.6\mathbb{P}(B), \qquad \text{so} \qquad \mathbb{P}(B) = \frac{0.55}{0.6} = \frac{11}{12} \approx \mathbf{0.9167}.$$

Independence forces a larger $\mathbb{P}(B)$ than mutual exclusivity does, because the overlap $\mathbb{P}(A, B)$ is now double counted in $\mathbb{P}(A) + \mathbb{P}(B)$ and has to be made up for.

2.2 Expectations and Variance

Each of the 3 repetitions draws a coin uniformly and flips it once, so a single trial gives heads with probability

$$\mathbb{P}(\text{heads}) = \frac{1}{2}(0.3) + \frac{1}{2}(0.9) = 0.6.$$

The coin is re-drawn on every repetition, so the three trials are independent and identically distributed Bernoulli(0.6) draws and $X \sim \text{Binomial}(3, 0.6)$. This is worth stating explicitly: X is *not* a mixture of a Binomial(3, 0.3) and a Binomial(3, 0.9), which is what it would be if one coin were chosen once and then flipped three times. The PMF is

$$\mathbb{P}(X = 0) = 0.4^3 = 0.064, \qquad \mathbb{P}(X = 1) = 3(0.6)(0.4)^2 = 0.288,$$

$$\mathbb{P}(X = 2) = 3(0.6)^2(0.4) = 0.432, \qquad \mathbb{P}(X = 3) = 0.6^3 = 0.216.$$

1. $\mathbb{E}[X] = np = 3(0.6) = \mathbf{1.8}$. Directly from the PMF, $0(0.064)+1(0.288)+2(0.432)+3(0.216) = 1.8$.

2. $\text{Var}[X] = np(1-p) = 3(0.6)(0.4) = \mathbf{0.72}$. Equivalently, $\mathbb{E}[X^2] = 0(0.064) + 1(0.288) + 4(0.432) + 9(0.216) = 3.96$, so $\text{Var}[X] = 3.96 - 1.8^2 = 3.96 - 3.24 = 0.72$.

3. $Y = \frac{1}{2+X}$ is a nonlinear function of X, so it has to be averaged against the PMF rather than evaluated at $\mathbb{E}[X]$:

$$\begin{aligned}\mathbb{E}[Y] &= \sum_{k=0}^{3} \frac{\mathbb{P}(X=k)}{2+k} = \frac{0.064}{2} + \frac{0.288}{3} + \frac{0.432}{4} + \frac{0.216}{5} \\ &= 0.032 + 0.096 + 0.108 + 0.0432 = \mathbf{0.2792} = \frac{349}{1250}.\end{aligned}$$

   For contrast, $\frac{1}{2+\mathbb{E}[X]} = \frac{1}{3.8} \approx 0.2632$, which is not the answer. The gap is Jensen's inequality: $1/(2+x)$ is convex on $x \geq 0$, so $\mathbb{E}[1/(2+X)] \geq 1/(2+\mathbb{E}[X])$.

2.3 A Variance Paradox?

There is no contradiction. The two formulas describe different situations, and the identity that reconciles them is

$$\text{Var}[U+V] = \text{Var}[U] + \text{Var}[V] + 2\text{Cov}(U,V).$$

The result $\text{Var}[X_1 + \cdots + X_n] = n\sigma^2$ is not a consequence of the X_i being identically distributed. It depends on independence, which forces every covariance term to vanish and leaves only the n variances. Writing $X + X$ does not produce two independent draws from F. It reuses one draw twice, so $\text{Cov}(X, X) = \text{Var}[X] = \sigma^2$ and

$$\text{Var}[X+X] = \sigma^2 + \sigma^2 + 2\sigma^2 = 4\sigma^2 = \text{Var}[2X],$$

which is the same answer the scaling rule $\text{Var}[aX] = a^2\text{Var}[X]$ gives. The apparent paradox comes from reading $X_1 + X_2$ and $X + X$ as the same expression. They are not: the first is a sum of two random variables that happen to share a distribution, the second is a deterministic transformation of a single random variable. Perfect correlation is the extreme case where the covariance term is as large as it can get, which is why $4\sigma^2$ exceeds the independent answer $2\sigma^2$.

I confirmed this numerically with 2×10^6 standard normal draws: the sample variance of X_1+X_2 for independent X_1, X_2 was 2.001, while the sample variance of $X_1 + X_1$ was 3.992, matching $2\sigma^2$ and $4\sigma^2$ respectively.

3 Calculus Review

3.1 One-variable derivatives

1. $f(x) = 3x^2 - 2x + 5$. Term by term, $\frac{d}{dx}(3x^2) = 6x$, $\frac{d}{dx}(-2x) = -2$, and the constant drops out, so $f'(x) = 6x - 2$.

2. $f(x) = x(1-x) = x - x^2$, so $f'(x) = 1 - 2x$.

3. $p(x) = \frac{1}{1+\exp(-x)}$ and $f(x) = x - \log(p(x))$. Two facts make this collapse. First, the logistic function satisfies $p'(x) = p(x)(1-p(x))$. Second, $\log p(x) = -\log(1+\exp(-x))$, so $f(x) = x + \log(1+\exp(-x))$. Differentiating the first form with the chain rule,
$$f'(x) = 1 - \frac{p'(x)}{p(x)} = 1 - \frac{p(x)(1-p(x))}{p(x)} = 1 - (1-p(x)) = p(x).$$
Using the hint $p(x) = 1 - p(-x)$, this can equivalently be written $f'(x) = 1 - p(-x)$. So f is the softplus function and its derivative is the logistic function, which is the relationship that makes the logistic loss convenient to optimize.

3.2 Multi-variable derivative

1. $f(\mathbf{x}) = x_1^2 + \exp(x_2)$. Each variable appears in one term, so
$$\nabla f(\mathbf{x}) = \begin{bmatrix} 2x_1 \\ \exp(x_2) \end{bmatrix}.$$

2. $f(\mathbf{x}) = \exp(x_1 + x_2x_3)$. With $u = x_1 + x_2x_3$ the chain rule gives $\partial f/\partial x_i = \exp(u)\partial u/\partial x_i$, and $\nabla u = [1, x_3, x_2]^\top$, so
$$\nabla f(\mathbf{x}) = \exp(x_1 + x_2x_3)\begin{bmatrix} 1 \\ x_3 \\ x_2 \end{bmatrix}.$$

3. $f(\mathbf{x}) = \mathbf{a}^\top\mathbf{x} = \sum_i a_ix_i$. Then $\partial f/\partial x_j = a_j$, so
$$\nabla f(\mathbf{x}) = \mathbf{a}.$$

4. $f(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x}$ with $A = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$. Writing it out as a summation,
$$f(\mathbf{x}) = \sum_i\sum_j x_iA_{ij}x_j = 2x_1^2 - 2x_1x_2 + x_2^2,$$
and differentiating each variable directly,
$$\nabla f(\mathbf{x}) = \begin{bmatrix} 4x_1 - 2x_2 \\ -2x_1 + 2x_2 \end{bmatrix} = (A + A^\top)\mathbf{x} = 2A\mathbf{x},$$
where the last step uses the fact that this particular A is symmetric. In general the gradient of a quadratic form is $(A + A^\top)\mathbf{x}$, and it simplifies to $2A\mathbf{x}$ only when $A = A^\top$.

5. $f(\mathbf{x}) = \frac{1}{2}\|\mathbf{x}\|^2 = \frac{1}{2}\sum_{i=1}^d x_i^2$. Then $\partial f/\partial x_j = x_j$, so
$$\nabla f(\mathbf{x}) = \mathbf{x}.$$
The factor of $\frac{1}{2}$ exists precisely to cancel the 2 from the power rule, which is why squared-error objectives are usually written with it.

4 Algorithms and Data Structures Review

1. Merge sort on n numbers costs $O(n \log n)$. The recurrence is $T(n) = 2T(n/2) + O(n)$, since the list is halved and the two sorted halves are merged in a single linear pass. There are $\log_2 n$ levels of recursion and each level does $O(n)$ merging work.

2. Finding the third-largest element of an unsorted list costs $O(n)$. One pass suffices while tracking the three largest values seen so far, which is $O(1)$ work per element. Sorting first would cost $O(n \log n)$ and does more than the question asks for. Note that the constant 3 matters here: the same argument gives $O(nk)$ for the kth largest, which is why this stays linear only because k is fixed.

3. Finding the smallest element greater than 0 in a *sorted* list of n numbers costs $O(\log n)$. Sortedness means the predicate "> 0" is false on a prefix and true on a suffix, so binary search finds the boundary in $O(\log n)$ comparisons.

4. Looking up the value for a key in a hash table with n numbers costs $O(1)$ on average. Hashing the key and indexing the bucket does not depend on n. The worst case degrades to $O(n)$ if every key collides into one bucket, but the expected cost under a good hash function is constant.

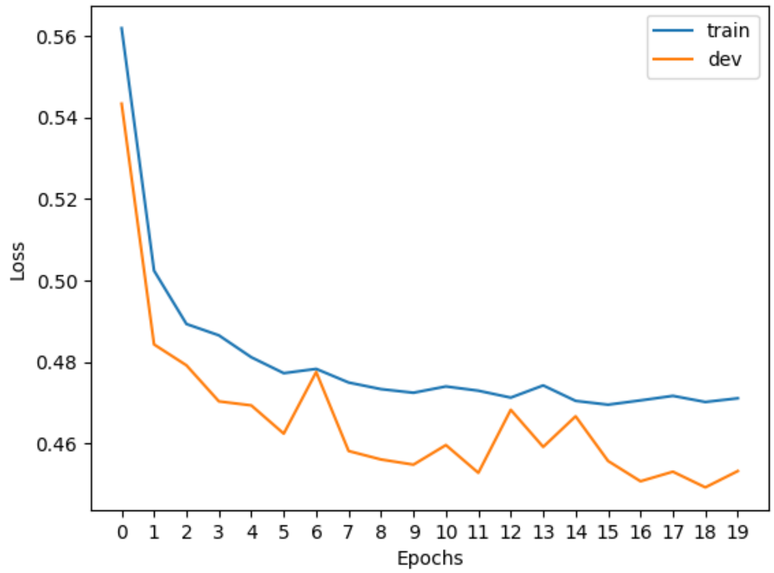
5. Computing $A\mathbf{x}$ with A of size $n \times d$ and $\mathbf{x}$ of size $d \times 1$ costs $O(nd)$. The result has n entries and each is a dot product of length d, giving nd multiply-adds.

6. Computing $\mathbf{x}^\top A\mathbf{x}$ with A of size $d \times d$ and $\mathbf{x}$ of size $d \times 1$ costs $O(d^2)$. Evaluate it as $\mathbf{x}^\top (A\mathbf{x})$. The inner matrix-vector product is $O(d^2)$ by the previous item with $n = d$, and the remaining dot product is $O(d)$, so the total is $O(d^2) + O(d) = O(d^2)$. The order of operations matters for memory rather than for the asymptotics: forming the outer product $\mathbf{x}\mathbf{x}^\top$ and taking its inner product with A does the same $O(d^2)$ arithmetic but needlessly allocates a $d \times d$ matrix.

7. Computing AB with A of size $m \times n$ and B of size $n \times d$ costs $O(mnd)$. The product has md entries and each is a dot product of length n. For square $n \times n$ inputs this is the familiar $O(n^3)$.

5 Programming

5.2.7 Run the pipeline: Train Loss vs. Dev Loss



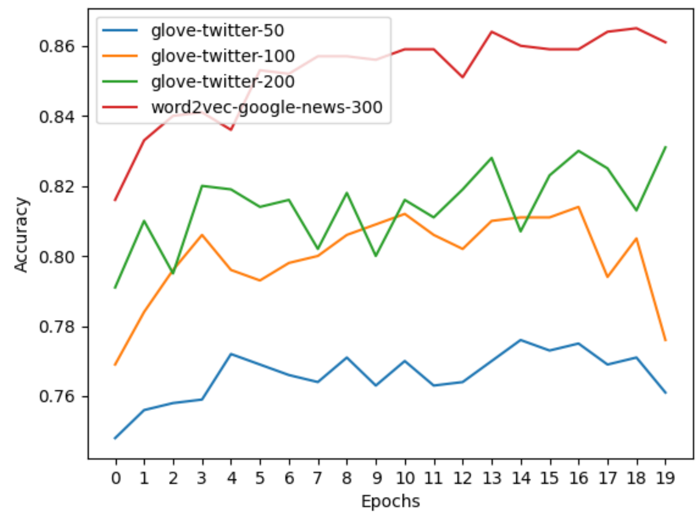
Configuration: `glove-twitter-50`, batch size 64, Adam at learning rate 0.025, 20 epochs. Selected values from the run:

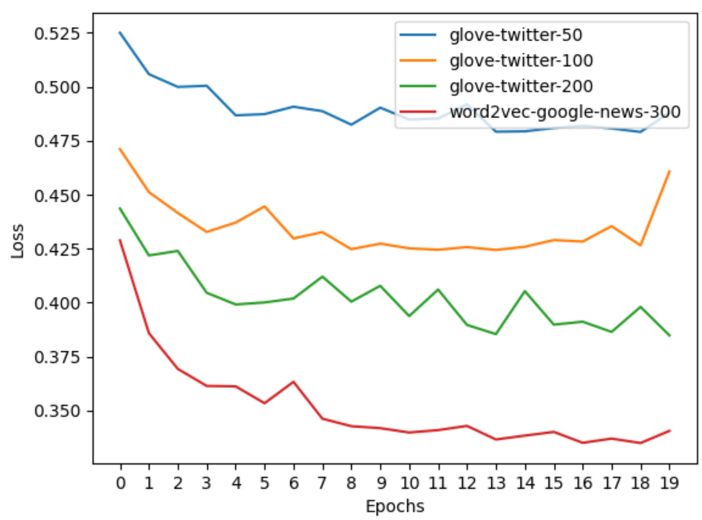
Epoch	0	5	10	15	17	19
Train loss	0.5619	0.4772	0.4740	0.4695	0.4717	0.4711
Dev loss	0.5434	0.4624	0.4596	0.4557	0.4531	0.4532
Dev acc	0.7140	0.7930	0.7850	0.7880	0.7940	0.7950

Test set: loss 0.4679, accuracy 0.7950. Minimum dev loss 0.4492 at epoch 18.

Findings. Train loss drops steeply for three epochs, then both curves flatten and the dev curve never turns upward, so the model is not overfitting: $50 \times 2 + 2 = 102$ trainable parameters against 20,000 examples. Dev loss settles below train loss, near 0.452 against 0.471, and over the final ten epochs its spread is four times as wide because it is measured on only 1,000 examples. The shared plateau is a limit of the representation rather than the optimizer, because averaging 50-dimensional word vectors discards word order and negation.

5.2.8 Run the pipeline: Explore Different Word Embeddings





Measured results across the four configurations, with vocabulary size and token coverage computed over a 500-review sample of the test split (139,530 tokens):

Embedding	Dim	Params	Vocab	Token cov.	Final dev loss	Test acc.
`glove-twitter-50`	50	102	1,193,514	97.8%	0.4880	0.772
`glove-twitter-100`	100	202	1,193,514	97.8%	0.4608	0.803
`glove-twitter-200`	200	402	1,193,514	97.8%	0.3848	0.821
`word2vec-google-news-300`	300	602	3,000,000	74.5%	0.3406	0.846

Findings. All three GloVe-Twitter runs share one vocabulary and identical 97.8% token coverage, so width is the only variable, and test accuracy climbs 0.772 to 0.803 to 0.821. Yet `word2vec-google-news-300` wins at 0.846 with only 74.5% token coverage, because it omits `a`, `of`, `and`, `to`, and all punctuation, which acts as a free stopword filter on the average. I would not credit that to width alone, though, since it also changes training corpus and algorithm.